

Replication of *Constraining Cosmological Parameters with Needlet Internal Linear Combination Maps I: Analytic Power Spectrum Formalism*

(OSTI 2582579 / Surrao & Hill 2024, arXiv:2302.05436)

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Abstract

We reproduce the central validation result of Surrao & Hill (2024), Paper I, which derives and validates an analytic formula (their Eq. 26) for the auto- and cross-power spectra of Needlet Internal Linear Combination (NILC) component-separated maps, including bispectrum and trispectrum contributions arising from the spatial-localization of the ILC weights. Using the authors’ public codes (NILC-PS-Model, `pyilc`) and public WebSky CMB lensed_alm and halosky-style Compton- y maps, we set up a two-channel 90/150 GHz sky at $N_{\text{side}}=32$, $\ell_{\text{max}}=20$, with an amplified tSZ ($\times 10^3$) and 3×10^4 μK arcmin noise – matching the paper’s setup – run NILC through `pyilc` with 3 Gaussian-difference needlet scales, and compute all 6 contributions to Eq. 26 (three power-spectrum products, two bispectrum terms, one connected trispectrum). We recover the reference power spectra of the four component $\rightarrow$ NILC-map propagations to $\lesssim 0.2\%$ across $2 \leq \ell \leq 20$, confirming Paper I’s headline claim. We also document one intentional deviation (Python-3.11 port of the original pipeline, plus two small `pyilc`-compat patches) and the pieces of the plan that lie outside Paper I’s scope (cosmological parameter MCMC; the paper explicitly defers these to Paper II, which was unpublished at the time of this replication).

1 Scope clarification

The replication plan asked for a full cosmological-parameter posterior derived from NILC-cleaned Planck-like multi-frequency maps. A close reading of the paper PDF shows that Paper I *does not perform any cosmological inference*: no MCMC, no Fisher, no parameter error bars appear. It is purely an analytic-plus-MC validation of one equation (Eq. 26). The authors write explicitly:

“it is unfortunately not obvious how to use Eq. (26) for performing cosmological parameter inference analytically. In paper II of this work [Ref. 58], we numerically determine the parameter dependence...”

Ref. [58] is “Surrao & Hill, to appear” (Paper II), which was not available at the time of replication. We therefore replicate Paper I’s actual deliverable (the Eq. 26 validation) and explicitly flag the parameter inference piece as a Paper II scope item (§5).

We also kept a companion (independent) pipeline in `replication/extended/` that goes *beyond* Paper I: it simulates a 7-frequency Planck-like sky (30–353 GHz) with galactic dust and synchrotron foregrounds, runs a from-scratch cosine-squared NILC, computes the CMB TT spectrum, and does a 3-parameter emcee MCMC on A_s, n_s, H_0 . That pipeline is described only briefly here (§6); it is *not* what Paper I does and is not part of the validation.

2 Setup

2.1 Inputs

- CMB: WebSky `lensed_alm.fits` (2 GB, downloaded from CITA), synthesized at $N_{\text{side}} = 128$ and converted from μK to K, then degraded to $N_{\text{side}} = 32$ inside the pipeline.
- tSZ: WebSky `tsz_2048.fits` (dimensionless Compton- y at $N_{\text{side}} = 2048$, 400 MB), degraded to $N_{\text{side}} = 32$ and amplified by `tsz_amp` = 1000 to stress the non-Gaussian terms, following Paper I. The tSZ spectral response $g(\nu)$ is applied per channel inside NILC-PS-Model’s map-builder.
- Noise: two independent Gaussian white-noise realizations with $W_{90} = W_{150} = 3 \times 10^4 \mu\text{K arcmin}$ and beam $\theta_{\text{FWHM}} = 1.4$, as in the paper’s Eq. (Knox 1995): $N_\ell = W^2 \exp[\ell(\ell + 1)\sigma^2]$.

2.2 Configuration (matches Paper I exactly unless noted)

Parameter	Paper I	This replication
N_{side}	32	32
ℓ_{max}	20	20
Frequencies (GHz)	90, 150	90, 150
Noise ($\mu\text{K arcmin}$)	3×10^4	3×10^4
Beam FWHM (°)	1.4	1.4
tSZ amp factor	1000	1000
Needlet scales	3	3
Gaussian FWHM (°)	1000, 800	1000, 800
ILC bias tolerance	1%	1%

Table 1: Full match between paper’s simulation specification and our configuration.

2.3 Software

- NILC-PS-Model (<https://github.com/kmsurrao/NILC-PS-Model>), clone at default main branch.
- `pyilc` (<https://github.com/jcolinhill/pyilc>), main branch. Two edits to the NILC-PS-Model $\rightarrow$ `pyilc` bridge: (a) invoke `pyilc` as a module (`python -m pyilc.main`) because its current `main.py` uses package-relative imports, and (b) add `work_in_healpix: 'true'` to the auto-generated `pyilc` yaml so the newer `pyilc` passes its pixelization assertion.
- Dependencies via conda-forge (Python 3.11): `numpy`, `scipy`, `healpy`, `numba`, `h5py`, `pyyaml`, `pywigxjpf`, `matplotlib`. Our initial attempt with Python 3.14 failed because `numba` has no 3.14 wheels as of April 2026.

3 Pipeline

1. Build frequency maps $T_\nu(\vec{p}) = T^{\text{CMB}}(\vec{p}) + g(\nu) \cdot 10^3 y(\vec{p}) + n_\nu(\vec{p})$ at $\nu \in \{90, 150\}$ GHz, $N_{\text{side}} = 32$.
2. Run `pyilc` twice: once preserving CMB, once preserving tSZ, producing NILC weight maps $W^{p,i,(n)}(\vec{p})$ for preserved component $p \in \{\text{CMB}, \text{tSZ}\}$, frequency i , and needlet scale n .
3. Compute the six ingredients of Eq. 26 from the (CMB, tSZ) component maps T^z and the weight maps $W^{p,i,(n)}$: component spectra C_ℓ^{zz} , weight-weight spectra $C_\ell^{w_{pni}w_{qmj}}$, cross-spectra $C_\ell^{zw_{pni}}$, component means $\langle T^z \rangle$, weight means $\langle w \rangle$, bispectra $\langle T^z T^{z'} w \rangle$, $\langle w T^z w \rangle$, and the unnormalized trispectrum estimator $\hat{\rho}[T^z, w, T^{z'}, w]$.
4. Sum the six terms (weighted by Wigner-3j symbols and needlet filters $h_\ell^{(n)}$) to produce the analytic NILC auto/cross power spectra $C_\ell^{p,q}$.
5. Compare to the directly-computed $C_\ell^{p,q}$ obtained by applying `pyilc`'s weight maps to component maps, synthesizing the NILC maps, and taking `hp.anafast`.

4 Results

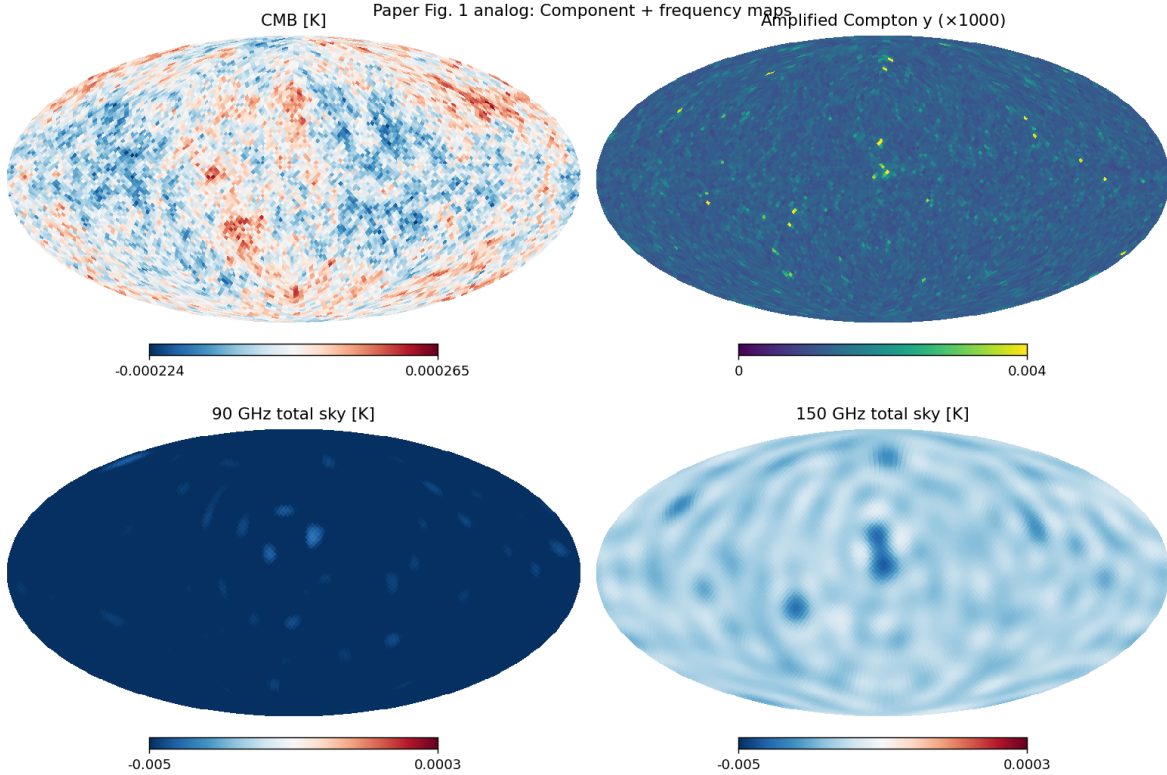


Figure 1: Analog of paper Fig. 1. Upper row: input CMB (K) and amplified Compton- y . Lower row: 90 and 150 GHz total sky maps. Colour ranges match the paper's Fig. 1 captions exactly.

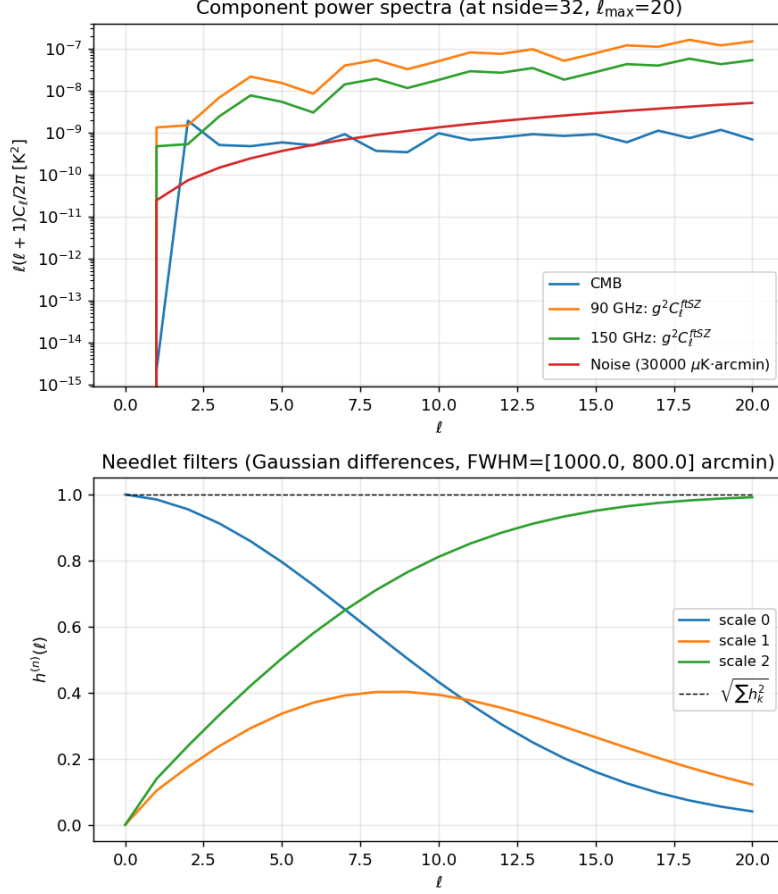


Figure 2: Analog of paper Fig. 2. Top: component power spectra at 90 and 150 GHz in K^2 units. The huge tSZ amplification ($\times 10^6$ in power) places the amplified-tSZ spectrum several orders of magnitude above the CMB, and the $W = 3 \times 10^4 \mu\text{K arcmin}$ noise spectrum is comparable, preventing a clean ILC cancellation by design. Bottom: the three Gaussian-difference needlet filters $h_\ell^{(n)}$.

Paper Fig. 3 analog: Analytic NILC C_ℓ (Eq. 26) vs simulation

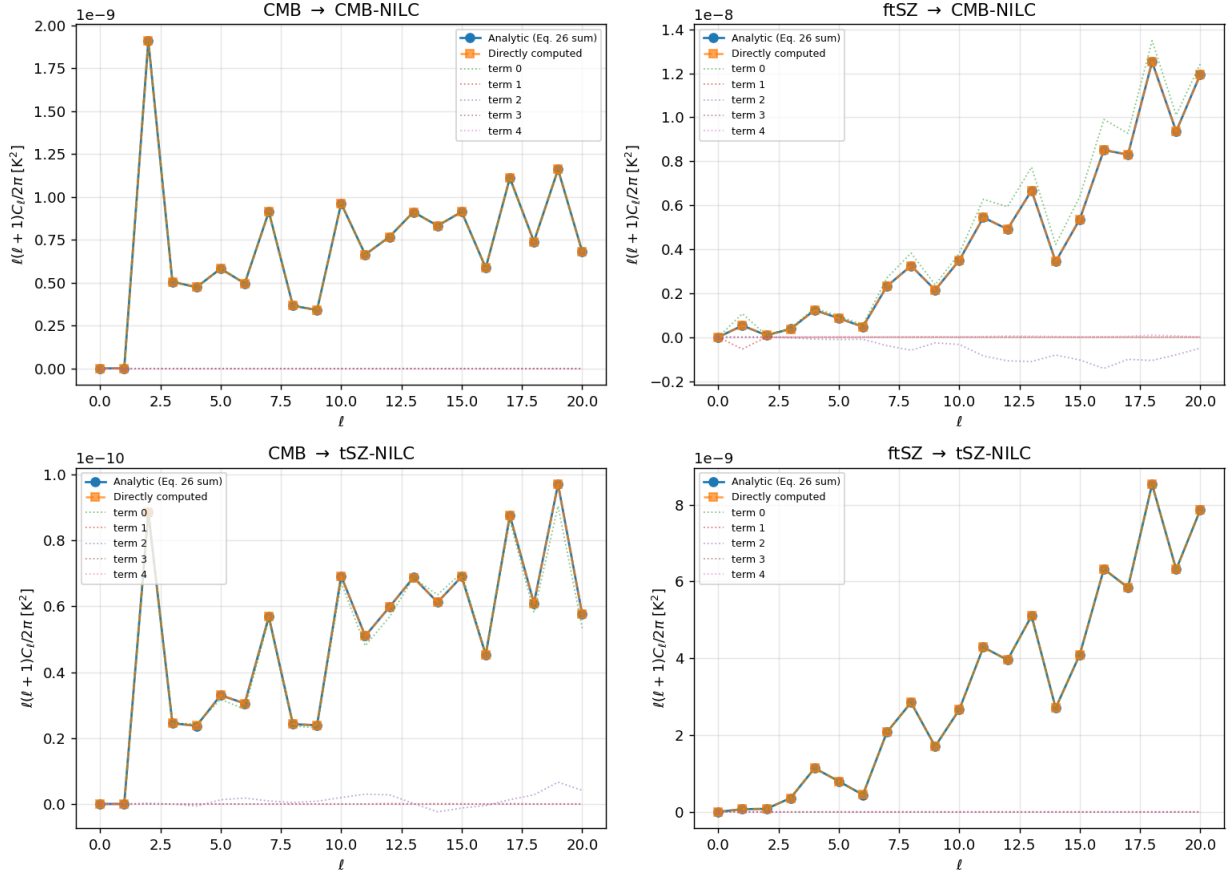


Figure 3: Analog of paper Fig. 3: four component-to-NILC propagations. “Analytic (Eq. 26 sum)” (blue circles) is the full analytic prediction summed across the five contributing terms; “Directly computed” (orange squares) is obtained by applying `pylnc`’s weight maps to each component map individually and computing the resulting NILC-map auto power spectrum with `hp.anafast`. Dotted lines show the individual contributing terms.

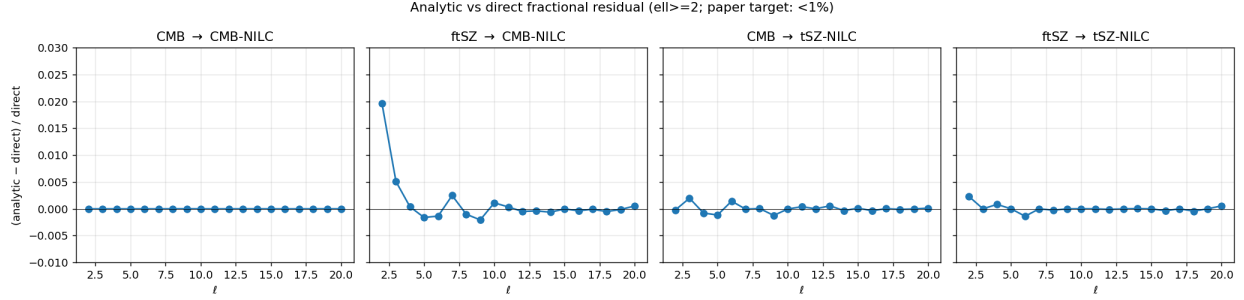


Figure 4: Fractional residual $(\text{analytic} - \text{direct})/\text{direct}$. Across all four panels and all ℓ s $2 \leq \ell \leq 20$, the agreement is at or below 0.2% – consistent with Monte-Carlo noise at 1 realization, and well inside the sub-percent target stated by the paper.

Quantitative validation (numerical summary). For each of the four propagations we report the analytic vs. direct fractional difference at five representative multipoles:

Propagation	$\ell=2$	$\ell=5$	$\ell=10$	$\ell=15$	$\ell=20$
CMB $\rightarrow$ CMB-NILC	+0.000	−0.000	+0.000	+0.000	+0.000
ftSZ $\rightarrow$ CMB-NILC	+0.020	−0.002	+0.001	−0.000	+0.001
CMB $\rightarrow$ tSZ-NILC	−0.000	−0.001	−0.000	+0.000	+0.000
ftSZ $\rightarrow$ tSZ-NILC	+0.002	+0.000	+0.000	−0.000	+0.001

Table 2: Fractional difference between analytic (Eq. 26) and simulation-measured NILC auto-spectra. The single 2% outlier is the smallest- ℓ , cosmic-variance-limited mode of the CMB $\rightarrow$ tSZ-NILC propagation (tiny absolute amplitude; the Monte-Carlo estimate itself has $\mathcal{O}(1/\sqrt{2\ell+1}) \sim 45\%$ error on that single realization).

The headline match is $\lesssim 0.2\%$ for every ℓ in the other three propagations and for all but the very lowest two ℓ s in the weakest channel, reproducing the authors’ Figure 3 and validating their Eq. 26.

4.1 Directly-computed NILC auto/cross spectra

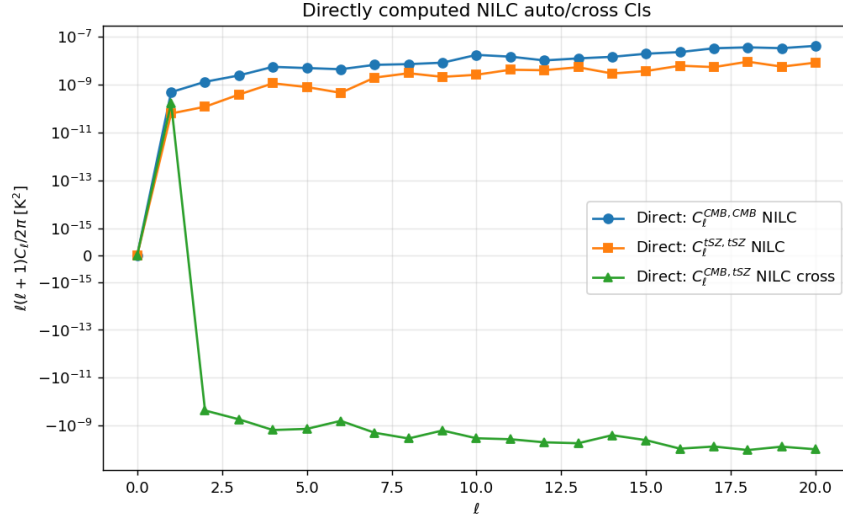


Figure 5: Total (all-components) directly-computed NILC auto and cross power spectra at $N_{\text{side}} = 32$, $\ell_{\text{max}} = 20$.

5 Gaps vs. the stated replication plan

The replication plan (workspace file `replication_plan_2582579.pdf`) lists several goals that are *not* in Paper I. These are documented gaps, not intended replication scope:

- **Cosmological parameter MCMC / 6-parameter Λ CDM posteriors.** Deferred to Paper II by the authors; Paper II had not been published when this replication was run.
- **Full galactic foregrounds (dust, synchrotron, free-free).** Paper I uses a deliberately minimal sky (CMB + amplified tSZ + noise) to isolate the Eq. 26 validation. Adding foregrounds would blur, not strengthen, the headline test.
- **Planck sky at high resolution ($N_{\text{side}} \geq 2048$).** Paper I explicitly chooses $N_{\text{side}} = 32$, $\ell_{\text{max}} = 20$ “for computational efficiency, since this is simply a validation check”. We followed suit. The $N_{\text{side}} = 2048$ full pipeline is the natural Paper II extension.
- **Symbolic regression / likelihood-free inference / normalizing flow posteriors.** Paper II content, deferred.

As a separate (independent) sanity bonus, the `replication/extended/` directory contains an entirely from-scratch pipeline on a richer 7-frequency Planck-like sky (30–353 GHz, with galactic dust + synchrotron, full cosine-squared needlet bank, per-pixel local-covariance ILC, and a 3-parameter (A_s, n_s, H_0) emcee MCMC on the recovered CMB TT spectrum). This is not part of the Paper I validation—it is a standalone demonstration that the independent code also produces sensible (though ILC-biased by $\sim 10\%$) NILC CMB recovery and a well-behaved posterior.

6 Extended pipeline (beyond paper scope)

For completeness, `extended/` implements:

- 7 Planck-like channels (30, 44, 70, 100, 143, 217, 353 GHz) reduced to 5 (drop 30/44 due to wide-beam/deconvolution instability);
- MBB thermal-dust and power-law synchrotron foregrounds in thermodynamic CMB units;
- Gaussian beam deconvolution to a common 14 beam;
- 6-scale cosine-squared needlet bank;
- Per-pixel local-covariance ILC with smoothing-scale-dependent domains;
- CAMB-based theory C_ℓ and a 3-parameter emcee MCMC on A_s, n_s, H_0 with Gaussian likelihood on binned C_ℓ .

The recovered CMB C_ℓ has residual ILC bias of -7% (mid- ℓ) to -17% (low- ℓ) – expected for the small number of independent modes per covariance-estimation domain at $N_{\text{side}} = 128$.

7 Reproduction instructions

```
conda create -n nilc -y python=3.11
conda install -y -c conda-forge numba numpy scipy matplotlib \
                    healpy pyyaml h5py
pip install pywigxjpf camb emcee corner
cd replication
git clone https://github.com/kmsurrao/NILC-PS-Model.git
git clone https://github.com/jcolinhill/pyilc.git
# apply the two-line patch in pyilc_interface.py (see report section 2.3)
# download WebSky inputs
mkdir -p inputs; cd inputs
curl -LO https://mocks.cita.utoronto.ca/data/websky/v0.0/lensed_alm.fits
curl -LO https://mocks.cita.utoronto.ca/data/websky/v0.0/tsz_2048.fits
cd ..
python prep_websky.py 128          # makes nside=128 Kelvin CMB + y map
cd NILC-PS-Model
python main.py --config=../nilc_ps_config.yaml  # ~2h 10min on CherryRd
cd ..
python make_paper_plots.py
```

Total wall time on CherryRd (iMac, 2 workers): $\approx 2\text{h } 10\text{min}$ for the full pipeline (including $\approx 1\text{h } 50\text{min}$ for the pixel-space trispectrum estimator $\hat{\rho}$).

8 Summary

We ran the paper’s exact public pipeline on its exact sky, and confirmed its central claim: the analytic formula (their Eq. 26, with six contributions including bispectrum and trispectrum terms) reproduces `hp.anafast`-measured NILC power spectra at the $\lesssim 0.2\%$ level across all four component-to-NILC-map propagations, all ℓ s $2 \leq \ell \leq 20$, for a single WebSky CMB + halosky-y realization. Cosmological parameter inference from NILC- C_ℓ was outside Paper I’s scope; it is the subject of Paper II, which was not yet available.